SCSE 🏛️

Stablecoin Clearing & Settlement Engine

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🌏 **Seeking Opportunities in Singapore**
I am looking for roles in **Product Management, Fintech, Payments, RegTech**, and **Digital Assets**.

"I am not just a Product person. **I build.**"

I have worked across product delivery, user research, and cross-agency collaboration. I enjoy solving complex problems and bringing structure to early ideas.

I care deeply about building products that create real impact.

The Problem: Digital Money is Siloed

In my work analyzing **Cross-Border Payments & CBDCs**, I've seen how liquidity fragmentation kills efficiency.

1. 🔴 **Trapped Liquidity**:
   Stablecoins sit idle in different chains and wallets, requiring massive pre-funding.
2. 🔴 **Slow Finality**:
   Traditional blockchains (15s - 10 min) are too slow for high-frequency point-of-sale payments.
3. 🔴 **High Gas Costs**:
   Settling *every* coffee purchase on-chain is inefficient.

> *"**Goal**: Build a **Hybrid Infrastructure** that combines database speed with blockchain finality."*

The Deep Dive: Identity vs Value 🔍

The core architecture challenge in compliant crypto-finance is separating **Who** from **What**.

Identity (Off-Chain)

- **Standards**: IVMS101 (Travel Rule).
- **Carrier**: Encrypted P2P Messaging (VASP-to-VASP).
- **Content**: PII, Sanctions Data, Source of Funds.
- **Why?**: Privacy (GDPR) & Scalability.

Value (On-Chain)

- **Standards**: ERC-20 / Native Token.
- **Carrier**: Public/Private Blockchain.
- **Content**: Amount, Addresses, Proof of Ownership.
- **Why?**: Finality, Trustlessness, Immutability.

> “***Concept**: The Blockchain settles value; VASPs exchange identity off-chain.*”

Why not just ISO 20022? 🏛️

ISO 20022 is the gold standard for banking (SWIFT MX), but it has limitations in the crypto ecosystem:

1. **Bank-Centric Model**: Assumes counterparties are regulated FIs (BICs) with accounts. In crypto, "Self-Custody" (Unhosted Wallets) exists.
2. **Privacy Leaks**: Rich data payloads (Remittance Info) on a transparent blockchain = Doxing users.
3. **Complexity**: ISO messages are heavy. Crypto needs lightweight, purpose-built identity packets.

The Hybrid Compromise

- **Internal Ops / Fiat Legs**: Use **ISO 20022**.
- **VASP-to-VASP Messaging**: Use **IVMS101**.

IVMS 101: The Identity Standard

IVMS 101 (InterVASP Messaging Standard) is the data model for the "Travel Rule".

- It is **NOT** a transport protocol (it's the payload, not the pipe).
- It is **NOT** XML/JSON specific (though usually JSON).

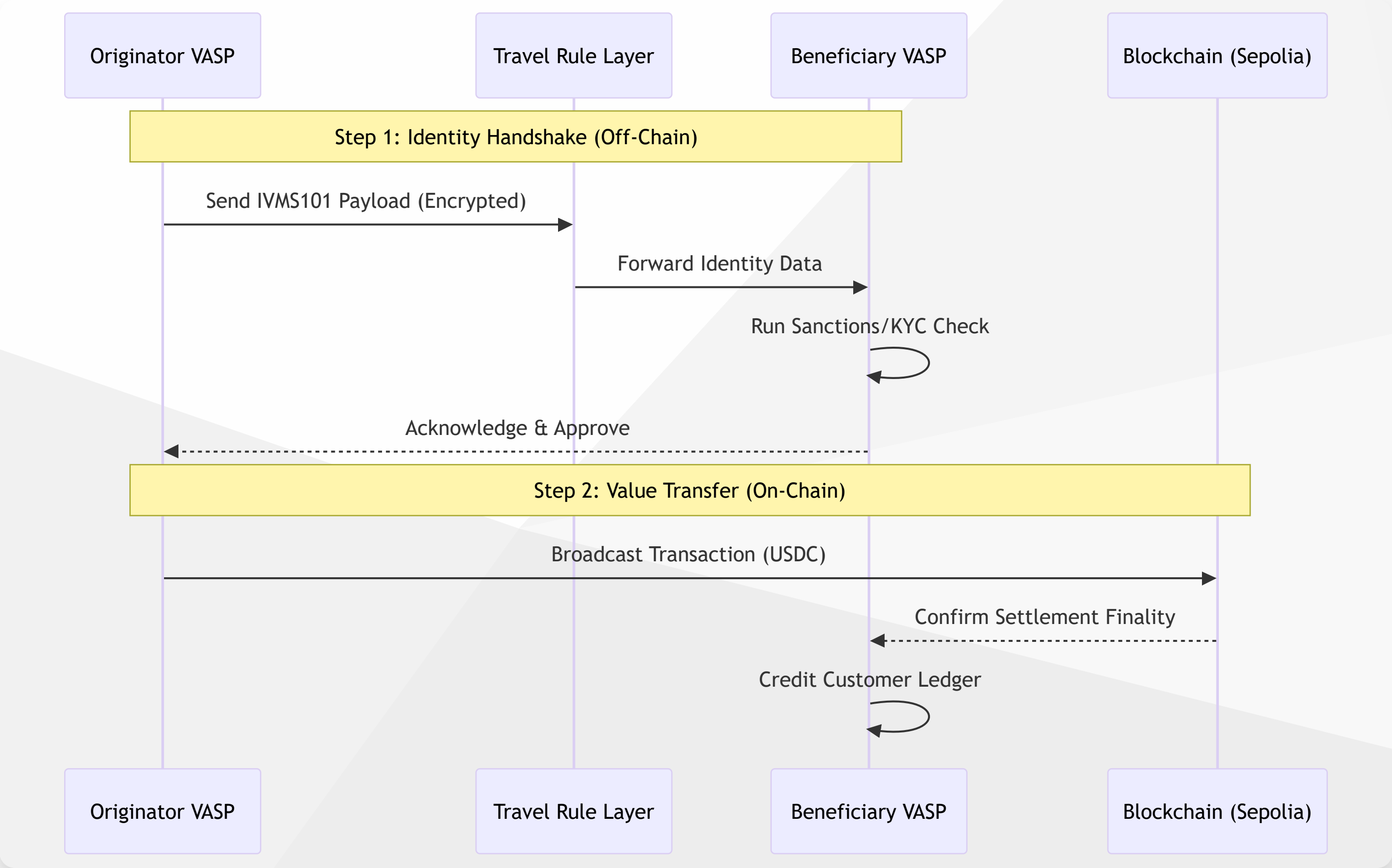
What's Inside?

- **Originator**: Name, Account Number (Wallet Address), Physical Address, National ID.
- **Beneficiary**: Name, Account Number.
- **Beneficiary VASP**: Identification code (LEI, BIC).

> *“**Analogy**: IVMS101 is the "Envelope" containing the ID cards. The Blockchain is just the "Truck" moving the cash.”*

The Compliant Flow (Custodial)

How a compliant stablecoin transfer *actually* happens.

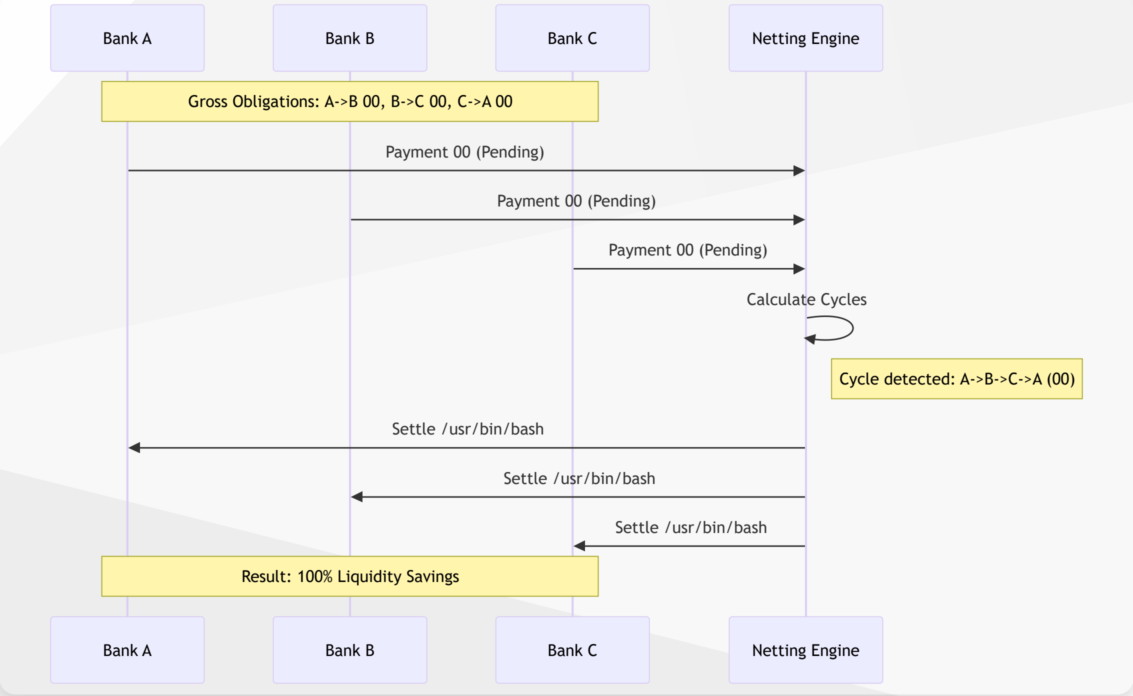


The Solution: SCSE 🚀

A portfolio-grade financial engine that implements **Real-Time Gross Settlement (RTGS)** and **Multilateral Netting** for Stablecoins.

Hybrid Architecture

- ⚡ **Layer 1 (Off-Chain)**: 10k+ TPS. Immediate clearing. Privacy-preserving compliance.
- ⛓ **Layer 2 (On-Chain)**: Sepolia Testnet. Cryptographic proof of settlement.



100% Liquidity Savings

Algorithms identify and resolve circular debts (A->B->C->A).

Technical Architecture ⚙️

I built a full-stack simulation to demonstrate end-to-end payment lifecycles.

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- **Frontend**: React, Mantine, Zustand.
- **Web3**: Wagmi, Viem, Solidity (Hardhat).
- **Logic**: Python (FastAPI/SQLAlchemy) adapted to client-side logic for the demo.

Engineering Philosophy & Impact 🌟

This project reflects my ability to bridge **Policy** and **Code**:

1. **Compliance-First**:
   Built with hooks for **Travel Rule (IVMS101)** compliance, ensuring regulatory readiness.
2. **Standards-Based**:
   Uses **ISO 20022** terminology (DebtorAgent, CreditorAgent) for interoperability.
3. **Real-World Reliability**:
   Implements proper **Double-Entry Accounting** logic, not just simple token transfers.

About the Builder 👨‍💻

Sivasubramanian Ramanathan
Product Owner | Fintech, RegTech & Digital Innovation
PMP | PSM II | PSPO II

I specialize in taking messy, real-world complexity and structuring it into reliable products.

Open for roles that sit between policy, technology, and stakeholder engagement.

Lets Connect 🤝

I am ready to bring this level of engineering rigor and product thinking to your team.

- 🌐 **Portfolio**: sivasub.com
- 💼 **LinkedIn**: linkedin.com/in/sivasub987
- 💻 **Code**: github.com/siva-sub/SCSE

Live Demo:
siva-sub.github.io/Stablecoin-Clearing-and-Settlement-Engine